

CUSTOMER CHURN DECISION DASHBOARD

A Business Intelligence Project

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Dataset: Telco Customer Churn (Kaggle)

Live Application: customer-churn-decision-dashboard.streamlit.app

Source Code: github.com/thanush912/customer-churn-decision-dashboard

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1. Introduction and Problem Statement

Customer churn — the decision by a subscriber to cancel their service — represents one of the most significant and measurable risks to a telecommunications business. When a customer leaves, the company loses not only the current month's revenue but also all future revenue that customer would have generated. Reactivating a lost customer is typically far more expensive than retaining an existing one.

Despite this, churn is often treated as an inevitable cost rather than a manageable risk. The core problem is informational: without a systematic way to identify which customers are likely to leave, when they are most at risk, and what factors are most strongly associated with their decision, retention resources cannot be deployed efficiently.

This project addresses that problem by building a three-page interactive business-intelligence dashboard for a telecommunications company, using the IBM Telco Customer Churn dataset. The dashboard translates raw customer data into clear KPIs, visual churn-driver analysis, a logistic regression risk-scoring model, and a set of data-backed recommended actions — all within a single, browser-based application ([live demo here](#)) that requires no specialist data-science knowledge to use.

2. Objectives and Business Goal

The primary business goal is to turn historical customer data into actionable decisions that reduce churn and protect monthly recurring revenue. The project was designed around four concrete objectives:

- **Visibility.** Surface the most important churn signals at a glance through clearly labelled KPI cards that a non-technical manager can read and act on immediately.
- **Diagnosis.** Identify which services, contract types, and payment methods are most strongly associated with churn, using bar charts and ranked tables computed directly from the data.
- **Prioritisation.** Score currently active customers by predicted churn probability using a logistic regression model, so the retention team can prioritise outreach to the highest-risk individuals.
- **Return on investment.** Quantify the estimated financial impact of specific retention actions — with explicit assumptions and adjustable scenario sliders — so decision-makers can evaluate ROI before committing resources.

Note: every finding in this report and in the dashboard describes an association observed in historical data, not a causal relationship. A correlation disclaimer is displayed on every dashboard page.

3. Dataset Description

The dataset used in this project is the IBM Telco Customer Churn dataset, made publicly available on Kaggle by IBM (kaggle.com/datasets/blastchar/telco-customer-churn). It contains records for 7,043 customers of a fictional telecommunications company, each described by 21 original features spanning demographics, subscribed services, contract terms, billing information, and a binary churn label. The dataset is a static snapshot — it does not include time-series behaviour — and is a widely used benchmark for churn-modelling exercises.

Table 1 — Dataset Summary

Property	Value
Total customers (rows)	7,043
Original features (columns)	21
Churned customers	1,869 (26.54%)
Active customers	5,174 (73.46%)
Data quality issue	11 blank TotalCharges values (all tenure = 0); fixed to 0.0

Table 2 — Key Columns

Column	Type	Description
customerID	String	Unique customer identifier
tenure	Integer	Number of months as a customer
Contract	Categorical	Month-to-month, One year, Two year
InternetService	Categorical	DSL, Fiber optic, No
PaymentMethod	Categorical	Electronic check, Mailed check, Bank transfer (auto), Credit card (auto)
TechSupport	Categorical	Yes, No, No internet service
MonthlyCharges	Float	Monthly bill amount in USD
TotalCharges	Float (cleaned)	Cumulative charges; 11 blanks fixed to 0.0
SeniorCitizen	Integer (0/1)	Whether the customer is a senior citizen
Partner / Dependents	Categorical	Whether the customer has a partner or dependents
Churn	Categorical	Target variable: Yes or No

The original dataset has 21 columns; a 22nd column, TenureGroup, was added during data preparation (Section 4.2). The gender column is present in the source data but was deliberately excluded from the predictive model — the rationale is explained in Section 4.3.

4. Data Cleaning and Preparation

4.1 TotalCharges Fix

The TotalCharges column was stored as a string rather than a numeric value, and eleven rows contained only whitespace. Inspection showed that all eleven affected rows had a tenure of 0 — customers who had just joined and had not yet been billed. The fix: strip whitespace, convert the column to a float using `pandas.to_numeric(errors='coerce')`, then fill the resulting NaN values with 0.0. This preserves all 7,043 rows while correctly representing zero charges for brand-new customers.

4.2 Added Columns

- **SeniorCitizenLabel** — a human-readable Yes/No label derived from the original 0/1 SeniorCitizen integer, used for charts and filters.
- **TenureGroup** — customers grouped into four bands (0–12, 13–24, 25–48, 49+ months) so bar charts and heatmaps can show how churn risk changes across the customer lifecycle without plotting 72 individual tenure values.

4.3 Gender Excluded from the Predictive Model

The gender column is present in the original dataset but was deliberately excluded as a model feature, for two reasons. First, gender showed negligible predictive value — churn rates for male and female customers are not meaningfully different. Second, and more importantly, using a protected demographic attribute to score customers for retention actions raises fairness concerns that outweigh any marginal accuracy benefit. The model relies entirely on service usage, contract terms, billing behaviour, and tenure — factors that are both relevant to churn and actionable by the business.

4.4 No Rows Removed

No rows were deleted during cleaning. There were no duplicate customerID values (verified by checking for any ID with a count greater than one), and the 11 TotalCharges blanks were fixed in place rather than dropped, since removing them would have discarded valid customer records.

5. Approach: From Data to Business Decisions

The project follows a structured business-intelligence flow, moving from raw data through progressively more actionable outputs:

- **Step 1 — Data.** The Telco CSV is loaded and cleaned as described in Section 4. Every number in the dashboard is computed at runtime; nothing is hard-coded.
- **Step 2 — KPIs.** Six summary metrics (churn rate, retention rate, monthly revenue lost, average tenure, average monthly charges, active customer count) are computed and displayed as KPI cards on Page 1.
- **Step 3 — Trends.** Churn rates are broken down by tenure group and contract type, showing that risk is highest in the first twelve months and among month-to-month customers.
- **Step 4 — Drivers.** Page 2 investigates which services, payment methods, and charge bands are most strongly associated with churn, including a ranked table of the five highest-churn customer segments (minimum 30 customers per group).
- **Step 5 — Risk scoring.** A logistic regression model is trained on 80% of customers and used to assign a churn probability to every active customer. Expected revenue at risk is computed as probability × monthly charge, summed across active customers — a financially meaningful estimate rather than a simple headcount.
- **Step 6 — Opportunities and actions.** Three retention scenarios (annual-contract conversion, auto-pay migration, tech-support upsell) are evaluated with an explicit, adjustable formula so a manager can explore scenarios without re-running any analysis.

Throughout, findings are described as associations observed in historical data, and a visible disclaimer reminds the reader that correlation does not imply causation.

6. System Design and Technology Stack

6.1 Single-File Architecture

The entire application — data loading, cleaning, model training, and all three dashboard pages — is contained in a single Python file, `churn_dashboard_app.py`. This was a deliberate design choice: it keeps the project easy to inspect, deploy, and hand over, with no separate training scripts, utility modules, or notebooks to manage.

6.2 Technology Stack

Layer	Library / Tool	Version Used
Front-end UI	Streamlit	1.35.0
Charts	Plotly	5.22.0
Data manipulation	pandas	2.2.2
Numerical computation	NumPy	1.26.4
Machine learning	scikit-learn	1.5.0
Programming language	Python	3.9 (tested)
Operating system (development)	Windows 10	—

6.3 Caching Strategy

Three caching decorators prevent redundant computation on every user interaction. `load_data` and `clean_data` are wrapped with `@st.cache_data`, so the cleaned DataFrame is only recomputed when the underlying file changes. `train_model` is wrapped with `@st.cache_resource`, which persists the fitted pipeline in memory for the server's lifetime, avoiding the cost of retraining on every page navigation.

6.4 Page Structure

Navigation between the three pages is handled by a sidebar radio widget. Page 3 additionally exposes four sliders: three that control the assumed conversion percentage for each retention action (5–50%, default 20%), and one that controls the high-risk probability threshold (0.30–0.80, default 0.50). Changing any slider triggers an immediate recalculation of the relevant estimates without reloading the page or the model.

7. Dashboard Walkthrough

The dashboard consists of three pages, navigated via a sidebar radio selector. It is deployed and publicly accessible at customer-churn-decision-dashboard.streamlit.app. Each page is described below alongside its screenshot.

7.1 Page 1 — Executive Overview

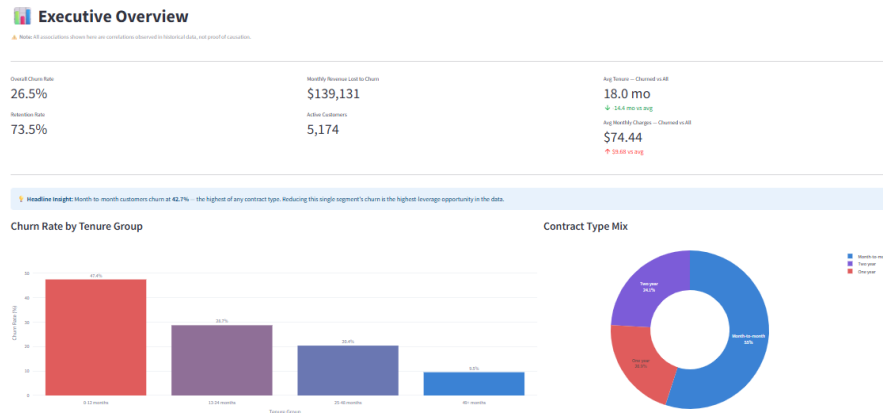


Figure 1: Executive Overview — KPI cards, churn by tenure group, and contract mix.

Page 1 opens with six KPI cards giving an immediate quantitative summary. The overall churn rate is 26.54% (retention rate 73.46%), and monthly revenue lost to churn — the sum of monthly charges across all churned customers — is \$139,130.85. The average tenure of churned customers (18.0 months) is notably shorter than the overall average (32.4 months), and churned customers carry a higher average monthly charge (\$74.44) than the overall average (\$64.76), showing that churn is disproportionately costly.

The bar chart below shows churn rate by tenure group: 0–12 months is highest at 42.7%, followed by 13–24 months (28.7%), 25–48 months (20.4%), and 49+ months (9.5%) — a clear gradient showing the first year is the most critical retention window. The donut chart shows the contract mix: 55% of customers are on month-to-month contracts, the largest and highest-risk group. A generated headline insight states: "Month-to-month customers churn at 42.7% — the highest of any contract type."

7.2 Page 2 — Services and Revenue Analysis

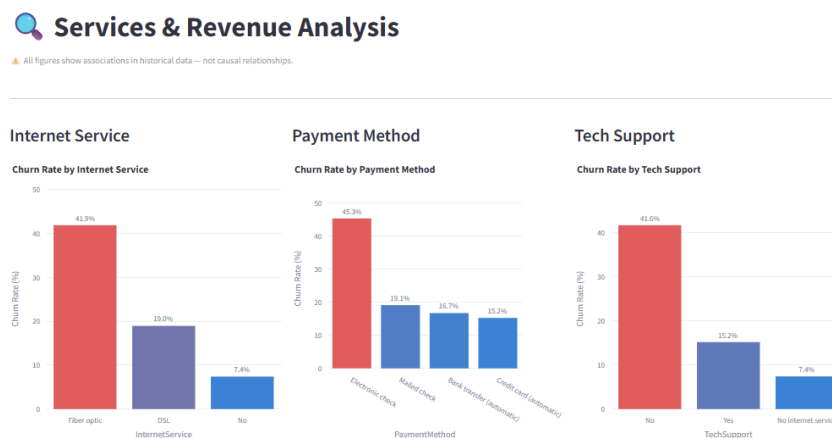


Figure 2: Churn rate by Internet Service, Payment Method, and Tech Support.

Page 2 investigates service-level drivers through three bar charts. Fiber optic customers churn at 41.9%, versus 19.0% for DSL and 7.4% for customers with no internet service. Electronic check customers churn at 45.3% — roughly three times the rate of automatic-payment customers (bank transfer 16.7%, credit card 15.2%). Customers without tech support churn at 41.6%, versus 15.2% for those who have it.

Monthly Charges Distribution — Churned vs Retained

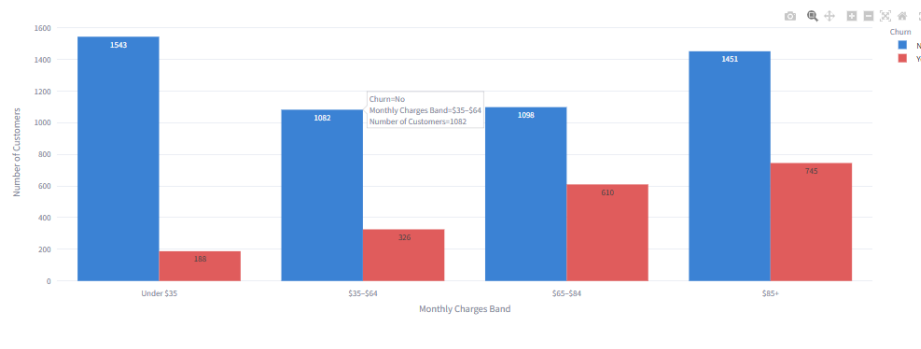


Figure 2a: Monthly charges distribution — churned vs. retained customers.

A grouped bar chart shows churned and retained customers across four monthly-charge bands: churn is most prevalent among customers paying \$65 or more per month.

Top 5 High-Churn Combinations (Contract × Internet × Payment Method)

Filtered to combinations with ≥ 30 customers for statistical reliability.

	Contract	InternetService	PaymentMethod	TotalCustomers	Churned	ChurnRate
1	Month-to-month	Fiber optic	Electronic check	1,307	789	60.4%
2	Month-to-month	Fiber optic	Mailed check	201	102	50.7%
3	Month-to-month	Fiber optic	Bank transfer (automatic)	327	149	45.6%
4	Month-to-month	Fiber optic	Credit card (automatic)	293	122	41.6%
5	Month-to-month	DSL	Electronic check	474	192	40.5%

Figure 2b: Top five highest-churn customer combinations (minimum 30 customers).

A ranked table identifies the five customer segments with the highest churn rates. The top segment — month-to-month contract, fiber optic internet, electronic check payment — churns at 60.4% across 1,307 customers, making it by far the highest-risk identifiable group in the data.

7.3 Page 3 — Risk, Opportunities and Recommended Actions

Risk, Opportunities & Recommended Actions

All churn associations are conditional. Revenue at Risk is a model estimate based on logistic regression probabilities.

Monthly Revenue Lost to Churn (Actual)
\$139,131

Estimated Revenue at Risk (Active Customers)
\$68,206

High-Risk Active Customers (≥ 50%)
533

Estimated: $\text{Monthly Churn Probability} \times \text{Monthly Charges}$ over all active customers.

Active customers with predicted churn probability ≥ 50%. Adjust the threshold slider to the slider.

Model Performance (Logistic Regression - Test Set)

Churn Rate Heatmap — Contract Type × Tenure Group

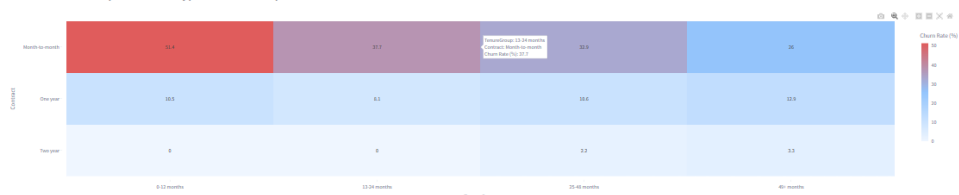


Figure 3: Revenue-risk KPIs and the churn-rate heatmap (contract type × tenure group).

Page 3 combines statistical findings with model-based scoring and explicit recommendations. The heatmap makes the highest-risk cells immediately visible — month-to-month customers in their first 12 months show the darkest shading. The model-based Revenue at Risk KPI, computed as each active customer's churn probability multiplied by their monthly charge and summed, is estimated at \$68,206 per month. A High-Risk Active Customers KPI counts active customers above a user-selected probability threshold (default 0.50), identifying 533 customers at that setting.

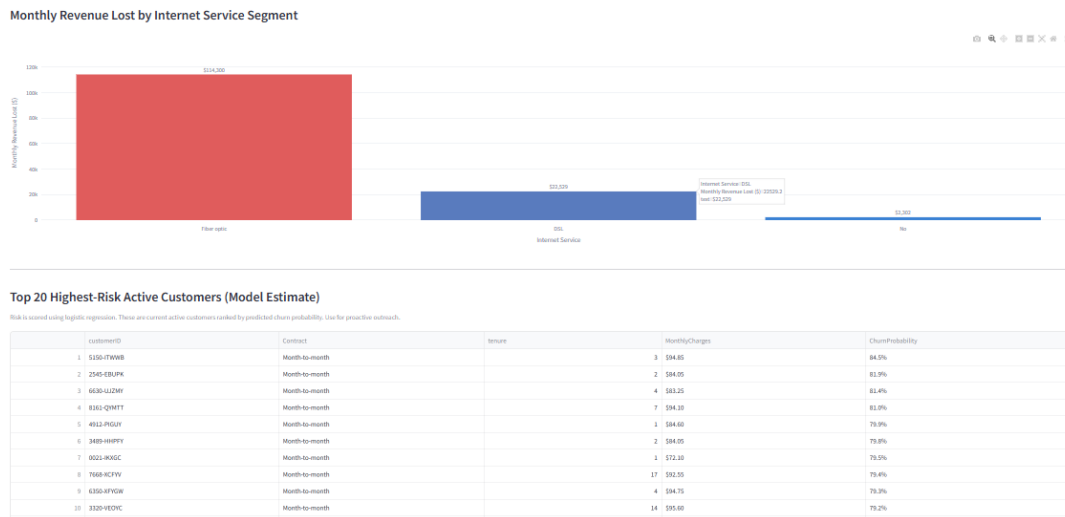


Figure 3a: Revenue lost by internet service segment, and the top-20 highest-risk active customers.

A bar chart breaks the \$139,130 monthly loss down by internet service: fiber optic accounts for \$114,300, DSL for \$22,529, and no-internet customers for \$2,302. Below it, a table lists the 20 active customers with the highest predicted churn probabilities — customer ID, contract type, tenure, monthly charge, and probability score — with a CSV export button so the retention team can push this list directly into a campaign system.

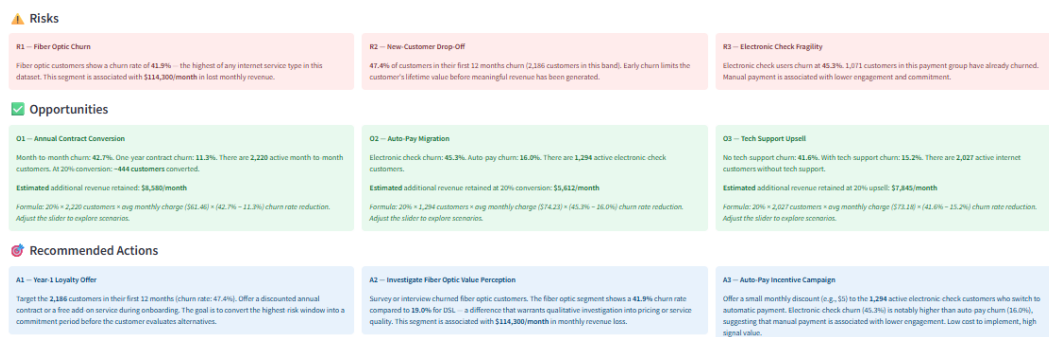


Figure 3b: Risks, Opportunities, and Recommended Actions cards.

The page closes with nine cards — three risks, three opportunities (each with an estimated financial impact and explicit formula), and three recommended actions — detailed in full in Section 10.

8. Churn Prediction Model

8.1 Method

The predictive model is a Logistic Regression classifier, chosen for its interpretability and well-calibrated probability outputs — both important in a business-intelligence context where scores must be explainable to non-technical stakeholders.

The model is trained on 18 features: 4 numeric (tenure, MonthlyCharges, TotalCharges, SeniorCitizen) and 14 categorical (Contract, InternetService, PaymentMethod, TechSupport, OnlineSecurity, OnlineBackup, DeviceProtection, PaperlessBilling, Partner, Dependents, PhoneService, MultipleLines, StreamingTV, StreamingMovies). Gender is excluded as described in Section 4.3, and customerID is retained only for identification in the output table, never as a model feature.

Preprocessing is handled inside a single scikit-learn Pipeline containing a ColumnTransformer: numeric features are standardised with StandardScaler, and categorical features are one-hot encoded with OneHotEncoder(handle_unknown='ignore') — the correct approach for nominal categories such as contract type and payment method, which have no natural ordering. The data is split into training (80%, 5,634 customers) and held-out test (20%, 1,409 customers) sets using a stratified split with random_state=42, so the proportion of churned customers is identical in both. The model is trained only on the training set; all metrics below are reported on the test set, which the model never saw.

8.2 Test-Set Performance

Metric	Value	Interpretation
Accuracy	80.6%	Correct predictions on the test set
Precision	65.7%	Of customers flagged as churners, 65.7% actually churned
Recall	55.9%	Of actual churners, 55.9% were correctly identified at threshold 0.50
ROC-AUC	0.842	Probability a random churner is ranked above a random non-churner

8.3 Recall Trade-Off

At the default 0.50 threshold, the model correctly identifies 55.9% of actual churners (recall), missing 44.1% of true churn cases, while precision of 65.7% means a flagged customer is correct roughly two-thirds of the time. This trade-off is acceptable for the use case: the model is not intended to make automatic decisions, only to help a human retention team prioritise outreach. An incorrectly flagged customer costs only the time of a retention call; a missed churner costs the company its future monthly revenue. The ROC-AUC of 0.842 confirms the model has meaningful discriminating power as a ranking tool even where recall at the default threshold is imperfect. The dashboard's adjustable threshold slider (0.30–0.80) lets the retention team trade recall for precision depending on campaign budget.

9. Key Findings

All figures below were computed directly from the dataset using pandas. As with every result in this report, these are correlational associations from a historical snapshot, not evidence of causation.

Table 5 — Key Churn Drivers

Driver	Segment	Customers	Churn Rate
Contract type	Month-to-month	3,875	42.7%
Contract type	One year	1,473	11.3%
Contract type	Two year	1,695	2.8%
Tenure group	0–12 months	2,186	47.4%
Tenure group	13–24 months	1,024	28.7%
Tenure group	25–48 months	1,594	20.4%
Tenure group	49+ months	2,239	9.5%
Internet service	Fiber optic	3,096	41.9%
Internet service	DSL	2,421	19.0%
Internet service	No internet	1,526	7.4%
Payment method	Electronic check	2,365	45.3%
Payment method	Mailed check	1,612	19.1%
Payment method	Bank transfer (auto)	1,544	16.7%
Payment method	Credit card (auto)	1,522	15.2%
Tech support	No tech support	3,473	41.6%

The most striking finding is the contract-type gradient: month-to-month customers churn at roughly 15 times the rate of two-year contract customers. The second strongest signal is tenure — customers who leave typically do so within their first 18 months (average tenure of churned customers: 18.0 vs. 32.4 months overall). The five highest-risk segments identified in Section 7.2 — all involving month-to-month contracts, fiber optic internet, and various payment methods — account for a disproportionate share of total churn volume.

10. Risks, Opportunities and Recommended Actions

All estimated revenue figures below use the same formula: *estimated revenue retained* = *conversion count* × *average monthly charge of segment* × (*churn rate before* – *churn rate after*). Estimates assume a 20% conversion rate (the dashboard default) and are clearly labelled as estimates; the dashboard sliders allow exploration between 5% and 50%.

10.1 Risks

- Risk 1 — Fiber Optic Churn.** Fiber optic customers churn at 41.9%, the highest of any internet service type (3,096 customers; 1,297 churned), associated with \$114,300 of monthly revenue loss — 82% of the total. The premium segment shows the strongest churn signal, suggesting a possible price–value mismatch or service-quality concern worth investigating.
- Risk 2 — New-Customer Drop-Off.** 47.4% of customers in their first 12 months churn (1,037 of 2,186). Early churn limits lifetime value before meaningful revenue has been generated; average tenure of all churned customers is only 18.0 months versus 32.4 months for all customers.

- **Risk 3 — Electronic Check Payment.** Customers paying by electronic check churn at 45.3% (1,071 of 2,365) — roughly three times the automatic-payment rate — suggesting manual payment may correlate with lower customer commitment or engagement.

10.2 Opportunities

- **Opportunity 1 — Annual Contract Conversion.** Month-to-month churn is 42.7% vs. 11.3% for one-year contracts. Of 2,220 active month-to-month customers, converting 20% (444 customers) at an average charge of \$61.46 and a 31.4-point churn reduction retains an estimated \$8,580/month.
- **Opportunity 2 — Auto-Pay Migration.** Electronic check churn (45.3%) versus auto-pay churn (~16.0%). Of 1,294 active electronic-check customers, converting 20% (258) at \$74.23 average charge and a 29.3-point reduction retains an estimated \$5,612/month.
- **Opportunity 3 — Tech Support Upsell.** Customers without tech support churn at 41.6% versus 15.2% with it. Of 2,023 active internet customers without tech support, upselling 20% (405) at \$73.18 average charge and a 26.4-point reduction retains an estimated \$7,845/month.

10.3 Recommended Actions

- **Action 1 — Year-1 Loyalty Offer.** Target the 2,186 customers in their first 12 months with a discounted annual contract or a free add-on during onboarding. The 47.4% first-year churn rate is the single most addressable signal in the data, and converting even a fraction of these customers to longer commitments before they evaluate alternatives is the highest-leverage action available.
- **Action 2 — Investigate Fiber Optic Value Perception.** Survey or interview churned fiber optic customers. The 41.9% churn rate (vs. 19.0% for DSL) warrants qualitative investigation into pricing, reliability, or expectations; this segment is associated with \$114,300/month in revenue loss.
- **Action 3 — Auto-Pay Incentive Campaign.** Offer a small monthly discount to the 1,294 active electronic-check customers who switch to automatic payment. Electronic-check churn (45.3%) notably exceeds auto-pay churn (~16%), and the intervention cost is low relative to the retention value.

11. Limitations and Future Work

11.1 Limitations

- **Correlation, not causation.** Every association in this report reflects patterns in historical data. That fiber optic customers churn more than DSL customers, for example, does not establish that the service type causes churn — confounding factors such as price, contract terms, or geography may explain part or all of the difference.
- **Snapshot data.** The dataset is a static snapshot with no date column, so there is no way to determine when customers joined or churned, or whether the observed patterns are stable over time. A live deployment would need a regularly refreshed data feed.
- **Estimated impact figures.** The opportunity estimates use a simplified formula (conversion rate × segment average charge × churn-rate differential) that does not account for campaign costs, response rates, cannibalisation of natural renewals, or systematic differences between customers who convert and those who don't.

- **Model recall trade-off.** At the default 0.50 threshold the model misses 44.1% of actual churners — a deliberate design choice for a prioritisation tool, but it means a meaningful share of high-risk customers will not appear in the top-20 list unless the threshold is lowered.
- **Fairness review needed.** Excluding gender is a first step, but a production deployment would need to audit model outputs across all protected demographic groups to ensure scoring does not systematically disadvantage any group.

11.2 Future Work

- **Live data integration** — connect the dashboard to a live customer database via API or scheduled ETL for real-time risk monitoring and an automatically refreshed at-risk list.
- **Advanced models** — gradient-boosted trees (XGBoost, LightGBM) typically achieve higher recall on churn datasets; a comparison study could test whether the added complexity is justified.
- **SHAP explanations** — adding SHAP values to the top-20 table would show which specific factors drive each customer's individual risk score, not just that they are at risk.
- **Campaign outcome tracking** — recording actual outcomes of triggered retention campaigns would allow the impact formulas to be calibrated against real results over time.

12. Conclusion

This project demonstrates that a well-designed business-intelligence dashboard can translate raw customer data into clear, evidence-backed decisions without requiring a specialist data-science background to interpret the outputs.

The analysis of the IBM Telco Customer Churn dataset shows that churn in this business is not spread evenly across the customer base — it is concentrated in three overlapping segments: month-to-month customers (42.7% churn), customers in their first twelve months (47.4%), and fiber optic users paying by electronic check (up to 60.4% in the highest-risk combination). Monthly revenue associated with churned customers is \$139,130.85.

The logistic regression model achieves 80.6% accuracy and a ROC-AUC of 0.842 on the held-out test set, making it a useful tool for ranking active customers by churn risk. Its expected-value approach to revenue-at-risk (\$68,206 estimated per month) gives a financially meaningful, probabilistic estimate rather than a binary oversimplification.

The three recommended actions — a Year-1 loyalty offer, a fiber optic quality investigation, and an auto-pay incentive campaign — are directly grounded in the data and estimated to retain a combined \$22,037 per month at the 20% conversion assumption. All estimates are clearly labelled, all assumptions stated, and the dashboard lets decision-makers adjust assumptions and see the impact immediately.

The project was built and tested on Python 3.9 using Streamlit 1.35, Plotly 5.22, pandas 2.2, NumPy 1.26, and scikit-learn 1.5. The complete application is contained in a single Python file and is publicly deployed at customer-churn-decision-dashboard.streamlit.app, with source code available on [GitHub](https://github.com).